

Research Overview: Lydia Project

Project subtitle:

Investigating Prefrontal–Premotor Network Activation During Piano Performance in Noise Using fNIRS

Objective:

To examine **prefrontal, temporal and sensorimotor cortical activity** and **sensory-motor area** during fine motor performance (piano playing) under two auditory conditions, quiet (baseline) and noisy (acoustic interference), using functional near-infrared spectroscopy (fNIRS).

A secondary aim is to include **a limited set of sensorimotor channels** to enable comparison with existing **finger-tapping data**, providing a reference for simple versus complex motor tasks.

A tertiary aim is to include physiological measurements (ECG) and task performance metrics to establish a mental workload (MW) and neural efficiency statistical framework suitable for future studies. The literature firmly establishes that there is no linear relationship between brain activity and task performance, thus two individuals with the same task performance can have significant differences in brain activity. By combining fNIRS, HRV, and task performance, we can map a complete picture of the individual player's response to auditory noise.

Rationale:

Piano playing is a complex sensorimotor task requiring precise timing, auditory feedback, and executive control. This involves coordinated activity between:

- **Prefrontal cortex (executive control, attention)**
- **Premotor cortex (motor planning)**
- **Supplementary motor area (motor sequencing and timing)**
- **Sensory-motor area (feedback of the mechanosensory, proprioception)**
- **Temporal lobe (auditory processing, sensorimotor integration, language)**

Environmental noise may disrupt auditory-motor coupling, increasing reliance on **prefrontal cortex (PFC)** and modulating **motor planning networks (SMA/PMA)**.

fNIRS is well-suited to capture these dynamics due to its quiet operation in upright positions such as piano playing, robustness to motion and sensitivity to **frontal, medial and temporal cortical activity** in ecologically valid tasks.

Link to Existing Work:

This protocol builds on preliminary **finger-tapping fNIRS data**, allowing:

- Comparison of **simple motor execution (tapping)** vs. **complex sequential performance (piano)**
 - Evaluation of **increased SMA/PMA involvement** with task complexity
 - Exploratory comparison of **sensorimotor activation patterns**
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Hypotheses:

- Piano playing in noise will elicit greater activation in:
 - **Dorsolateral prefrontal cortex (DLPFC)**
 - **Premotor cortex (PMA)**
 - **Supplementary motor area (SMA)**
compared to quiet conditions.
 - Increased oxyhemoglobin (HbO) response will correlate with perceived task difficulty and timing variability.
 - Compared to finger tapping:
 - Piano playing will show **stronger SMA and PMA engagement**
 - **Prefrontal involvement will be increased**
 - Sensorimotor activation will be present but less dominant
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Experimental Protocol: Block Design fNIRS Study

Participants

- **N = 3 (pilot study, within-subject design)**
- Healthy adult pianist (right-handed, ≥5 years of formal training)
- Age: Adults
- Normal hearing, no neurological disorders
- Informed consent obtained

Apparatus

- fNIRS system (e.g., NIRXport2, 32 optodes, 760 & 850 nm)
 - Coverage focused on:
 - **Prefrontal cortex**
 - **Premotor cortex (FC region)**
 - **Supplementary motor area (midline: Fz/Cz)**
 - Includes **a small number of sensorimotor channels (C3/C4 region)** for comparison
 - Accelerometer enabled in Aurora and attached to cap.
 - 61-key MIDI keyboard (portable, velocity-sensitive)
 - Noise delivery system: headphones or loudspeakers
 - Background noise: multi-talker babble (~65 dB SPL)
 - Quiet condition: <30 dB SPL and non-reflective walls????
 - MIDI logging for accuracy and timing
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Experimental Design

- **Within-subject block design (pilot)**

Conditions:

- Natural room Piano Playing (QP)
 - Quiet environment Piano Playing (QP)
 - Noisy environment Piano Playing (NP)
- Rest (R)
- AA adjusted room Piano Playing (NP)
 - Quiet environment Piano Playing (QP)
 - Noisy environment Piano Playing (NP)
- Rest (R)

Each condition is repeated multiple times to allow for a **within-session comparison**.

Piano playing

For a strong response in the auditory cortex, we need a stimulus frequency > 1.5 Hz, which means the piano should be played at > 90 BPM (Binder 1994).

The player must fix their gaze on an agreed-upon point to avoid significant head tilts.

This also means the piece must be played without seeing their fingers.

Baseline Protocol

1 minute before the playing starts.

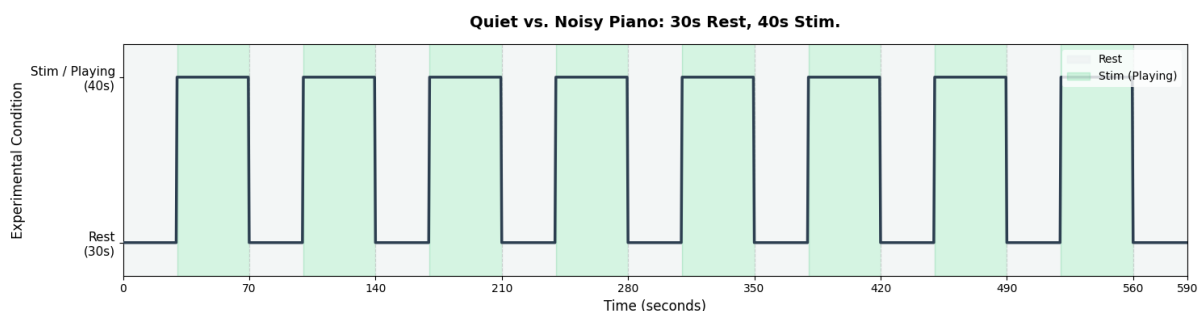
Block Design

Note: We need to randomize and stagger the block design to reduce habituation and improve statistical significance.

Each block (70 s) which is repeated 6 times?

- 40 s (+/- <8 seconds) – Active piano playing
- 30 s (+/- <5 seconds) – Rest (fixation, no movement)
- QP and NP is randomly alternated (50/50).
- Counterbalanced across blocks (as feasible within a single session).

Total duration: (~ 8 minutes + 2 min rest) $\times 3 = 30$ min.



Task Instructions

Participants perform a simple, memorized 8-bar melody:

- Erik Satie Gymnospedie no.1 was discussed as a simple and repeating piece.

- Example: broken chords or structured scale pattern that lasts between 30-40 seconds
- Must be played >90 BPM.
- Practiced before recording in both conditions.

During rest:

- Hands on keyboards
 - Eyes fixated on the cross
 - Breath normally
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Data Acquisition

fNIRS montage (hva syns du Adam?)

- **Spot montage (PFC, Temporal, SM)**
- Sampling rate: Max? Spot montage will reduce optode cross-talk
- No short channels at this level

Physiology via WINGS2:

- 3-lead ECG preferably on the torso.
- Respiration band (Can we regress the respiration from fNIRS signal?)

Data Analysis Plan

Quality Control

- Heartbeat
- Scalp coupling index

Preprocessing

- Motion correction (her burde vi kanskje ha med IMU på hodet til forsøkspersonen).
- Removal of systemic components and baselines
- Frequency-time wavelet analysis

- (Short channel regression, only applicable for HD-fNIRS)
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Analysis

Heart Rate Variability

HRV is calculated from the ECG data and compared against blocks and the HRV-HbO relation is used to gauge neural efficiency in quiet vs noisy conditions. We use 3 HRV metrics for statistics:

- **SDNN**: Standard deviation of NN interval time series (ms)
- **RMSSD**: Root mean square of successive differences between normal heartbeats (ms)
- **pNN50**: Percentage of successive RR intervals that differ by more than 50 ms (%)

Task Performance

We need to establish some sort of correctness measure, are the pressed notes correct and are they in time?

General Linear Model

Block-based GLM reveals which condition produced the strongest hemodynamic activation i.e. brain activity:

- QP
- NP

Block Averaging

Block-based averaging (BA) reveals which condition produced the strongest hemodynamic activation i.e. brain activity:

- QP
- NP

Diffuse Optical Tomography

Diffuse optical tomography (DOT) excludes extracerebral activity and reveals which condition produced the strongest hemodynamic activation i.e. brain activity:

- QP
- NP

GLM, BA and DOT are sides of the same coin, there should be small discrepancies between the results. NIRWizard handles all these analyses natively with automatic reporting and visualization.

Functional Connectivity

Functional connectivity (FC) is calculated by Pearson's r-value is calculated for each channel pair, the correlation matrix is Fisher's Z-transformed, and reveals how intercerebral communication differ between cases:

- QP
- NP

Regions of Interest

- **Primary ROIs:**
 - Prefrontal cortex (DLPFC/VLPFC)
 - Sensorimotor area (M1, S1, SMA, PMA)
 - Temporal lobe (T5/T7?)
- **Secondary (comparative):**
 - Sensorimotor cortex (C3/C4)

Pilot-Level Analysis Strategy

- Compare **QP vs NP within subject**
- Examine:
 - HbO and HbR changes across blocks
 - Consistency across repetitions
 - The area under the curve of frequencies
- NIRWizard test-case: a complete end-to-end data analysis with reported results never exiting NIRWizard

- Combine physiological (HRV) and task performance (% of notes correct and timings) statistics to gauge mental effort (mental workload) and neural efficiency. This is a methodo

Expected Outcomes

- Increased HbO and decreased HbR in the **prefrontal cortex during noise**
- Enhanced **SMA and PMA recruitment** under noisy conditions
- Clear difference between **simple (tapping) and complex (piano) motor networks**
- Modest but interpretable **sensorimotor activation**

Referanser:

J.R. Binder, S.M. Rao, T.A. Hammeke, J.A. Frost, P.A. Bandettini, J.S. Hyde, Effects of stimulus rate on signal response during functional magnetic resonance imaging of auditory cortex, *Cognitive Brain Research*, Volume 2, Issue 1, 1994, Pages 31-38, ISSN 0926-6410, [https://doi.org/10.1016/0926-6410\(94\)90018-3](https://doi.org/10.1016/0926-6410(94)90018-3), (<https://www.sciencedirect.com/science/article/pii/0926641094900183>), Keywords: Cerebral blood flow; Cerebral metabolism; Auditory cortex; Functional imaging; Speech perception; Temporal lobe.

[Laura B. Carlton](#), [Miray Altinkaynak](#), [Shannon Kelley](#), et al. "Surface-based image reconstruction optimization for high-density functional near-infrared spectroscopy," *Neurophotonics* 13(2), 025001 (14 Mar 2026) <https://doi.org/10.1117/1.NPh.13.2.025001>
